

Machine Learning Foundations

General Information, HSG

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The general “Course and Examination Fact Sheet” provides the legal basis of the course and contains a description of the course’s content together with information on the grading components.

The course is given in the form of 6 weekly sessions of 180 min each. Typically (apart from the first session), this will consist of the interactive lecture part + discussion of your solutions to the exercises and time to coordinate and reflect on the exercises for the following week.

Contact

You can contact me through Canvas or by sending an e-mail to lyudmila.grigoryeva@unisg.ch.

Office hours are held on request (only if your questions cannot be discussed in the breaks during the lectures).

In case of questions concerning the exercises, please contact Mr. Jonas Huwyler via jonas.huwyler@unisg.ch.

Content

- **Statistical Learning** with a focus on **supervised learning**: general problem structure and a revision of regression methods from this perspective.
- **Classification algorithms**: discriminant analysis, kernel density classification and naive Bayes classifier, logistic regression based classification, nearest neighbor classification, support vector machines, trees, neural networks; methods for combining regression and classification procedures through bagging, wisdom of crowds, or boosting.
- **Reinforcement learning**: multi-armed bandit problems, regret, algorithms based on upper confidence bounds and their optimality.
- **Unsupervised learning**: clustering (k-means and mixture models) and dimension reduction (principal

component analysis) will be covered if the time permits; it will be announced in advance whether this part of the course will be examined.

The main sources of inspiration for this course are:

- Richter [2019] (recommended, relatively inexpensive, perhaps a bit too technical, in German),
- Hastie et al. [2009] (available online, comprehensive, non-technical).

I also recommend:

- Zhou [2021] as an additional reference, and
- Lattimore and Szepesvári [2020] as a comprehensive reference for students interested in multi-armed bandits.

Other literature suggestions are provided in the course factsheet.

Course material

Lecture notes: Comprehensive weekly handouts are made available online (Canvas). They build the basis for the exam.

Slides: A subset of the lecture notes that serve the sole purpose of guiding the presentation/discussion in class.

Grading

The grading consists of two parts (the exercises carrying a weight of 2/5, the final exam a weight of 3/5):

- **Exercises**, which can be solved in groups of 2 (problem sets are due weekly from week 2 until week 5; each a maximum of 10 points, gives a total of 40 points)
- **Final exam** (week 6 (potentially at the time of the lecture); 90 minutes; short and long(er) questions; maximum of 60 points)

In total this gives a maximum of 100 points.

Exercises

The exercises (mostly computational ones in **python** or in **R** by the students' preference) are contained in the handouts and will be posted on Canvas as assignments.

To obtain points for a problem the following must be done:

- A solution (which can be the same for all members of a team, but every member of the team must upload) has to be submitted in written form (preferred is jupyter notebook and a pdf, but use what works best for you and produces a comprehensible result) on Canvas prior to the deadline, mentioning also the name of the other team member, if applicable.

Note: Every team has to be able to explain their submissions in a convincing manner together with answering clarifying questions appropriately. Typically, at least one member of the group should be present in the part of the lecture devoted to exercises, but extra appointments can be arranged (upon request per e-mail prior to the respective lecture) if this should not be possible. This format will be possibly changed depending on the number of participating students.

Final exam

Short questions: could be Single/Multiple choice, TRUE/FALSE, definitions, terminology, naming methods and important python functions, ...

Longer questions: explain a method, explain the differences between algorithms, explain a problem (e.g., classification) together with one algorithm to solve that problem, etc. These questions assess your level of comprehension of the material.

References

- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer: New York, 2009.
- Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.
- Stefan Richter. *Statistisches und maschinelles Lernen*. Springer: Berlin, 2019.
- Zhi-Hua Zhou. *Machine Learning*. Springer: Berlin, 2021.